

AI Resume Screening System

A Machine Learning and NLP based project that automatically screens resumes and classifies candidates based on skills, experience, and job role requirements.

Project Objective

To automate the resume shortlisting process using Natural Language Processing (NLP) and Machine Learning techniques.

Features

- Resume parsing and keyword extraction
- NLP-based text preprocessing
- Candidate classification using ML algorithms
- Skill matching and ranking system
- Easy integration with recruitment workflow

Technologies Used

Python, Pandas, NumPy, Scikit-learn, NLP, TF-IDF Vectorization, Jupyter Notebook

Machine Learning Workflow

1. Data Collection
2. Text Cleaning & Preprocessing
3. Feature Extraction using TF-IDF
4. Model Training
5. Prediction & Evaluation

Algorithms Used

Naive Bayes, Logistic Regression, Random Forest

Model Evaluation

Accuracy Score, Precision, Recall, F1 Score

Future Improvements

- Deep Learning integration
- Resume ranking dashboard
- Real-time recruiter analytics

Module	Description
Data Preprocessing	Cleaning and tokenization of resume text
Feature Engineering	TF-IDF Vectorization
Classification	Candidate role prediction
Evaluation	Accuracy and performance metrics

Created By: Ashish Khakre